

VIJITH V T

Applied AI Engineer | Python, RAG & LLM Applications | FastAPI Backend Engineering

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PROFESSIONAL SUMMARY

Software engineer with an M.Sc. in Artificial Intelligence, applying a backend-engineering foundation to Generative AI and LLM application development. Builds Python/FastAPI services integrating retrieval-augmented generation (RAG), embeddings, vector search and LLM APIs into practical applications. Hands-on experience spans resume/document processing, semantic and lexical matching, and prototype-level agentic AI workflows using LangGraph and MCP. Combines this applied AI/backend work with technical mentoring, reviewing engineers' and trainees' Python, automation and AI projects.

TECHNICAL SKILLS

AI / GenAI: Generative AI, LLM Applications, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Semantic Search, Vector Search

AI / Retrieval Tools: ChromaDB, Sentence Transformers, Groq LLM API

Backend Engineering: Python, FastAPI, REST APIs, MySQL, JWT Authentication, Role-Based Access Control (RBAC)

Engineering & Tooling: Docker, Docker Compose, Git

Prototype / Working Exposure: LangChain, LangGraph, AI Agents, Tool Calling, Model Context Protocol (MCP), Flask, Neo4j, Celery, Redis

Automation: Robot Framework, Selenium, Playwright, Appium

PROFESSIONAL EXPERIENCE

ThinkPalm Technologies Pvt. Ltd. — Associate Specialist, Academy & Research Oct 2022 – Present

- Develop Python/FastAPI backend services and prototype Generative AI workflows, including RAG pipelines, LLM API integration and embedding-based retrieval, for internal applied-engineering and learning use cases.
- Build and maintain automation scripts and test workflows using Robot Framework, Selenium, Playwright and Appium to support engineering validation processes.
- Prototype agentic AI proof-of-concepts using LangGraph and Model Context Protocol (MCP) to evaluate tool-calling and multi-step workflow patterns.
- Mentor engineers and trainees through technical sessions and hands-on projects covering Python, Generative AI, RAG, FastAPI and automation, reviewing implementation quality and architecture.
- Evaluate trainee and project submissions, providing feedback on code quality, design decisions and testing approach as part of engineering enablement.

Kerala State Centre for Advanced Printing & Training — Guest Faculty, Computer Science Sep 2018 – Oct 2022

- Taught core computer science and programming concepts (C, Python, databases) to students, connecting theoretical fundamentals to practical, hands-on exercises.
- Guided student projects and coursework, supporting problem-solving skills and introducing emerging areas including AI and automation.

Earlier Career (2015 – 2020): Program Assistant, Central Adoption Resource Authority | Project Assistant, National Health Mission | Commercial Apprentice, Kerala State Pollution Control Board — administrative and project-support roles prior to transitioning into technical education and software/AI engineering.

SELECTED AI PROJECTS

TalentHunt — AI-Assisted Recruitment & Resume Matching Platform

- Designed and developed an AI-assisted recruitment platform to help recruiters review applicants and compare candidates against job descriptions, moving beyond manual keyword-based screening.
- Built a modular FastAPI backend with MySQL persistence and JWT-based authentication, implementing role-based recruiter and candidate workflows.
- Implemented resume text extraction and processing using PyMuPDF as the first stage of the candidate evaluation pipeline.
- Implemented semantic and lexical Resume-JD matching using Sentence Transformer embeddings, ChromaDB retrieval and RapidFuzz fuzzy matching, combining both signals into a hybrid matching approach.

- Integrated Groq-hosted LLM workflows to generate AI-assisted candidate evaluation output for recruiter review, and containerized the application (frontend, FastAPI backend, MySQL, Nginx) using Docker Compose.

Technologies: Python, FastAPI, MySQL, JWT, RBAC, PyMuPDF, Sentence Transformers, ChromaDB, RapidFuzz, Groq LLM API, Docker, React (frontend interface)

Engineering Knowledge Graph — AI/Backend Prototype

- Developed a backend prototype exploring how fragmented engineering information — repositories, Git activity, Jira items, commits and documentation — could be centralized into a searchable Neo4j graph.
- Built Python/Flask services with Celery and Redis for asynchronous processing of engineering data sources.
- Explored AI-assisted repository and engineering-knowledge analysis workflows on top of the graph structure.

Technologies: Python, Flask, Neo4j, Celery, Redis, Docker

Agentic AI & MCP Prototypes

- Built prototype AI agent workflows exploring tool-calling and multi-step task orchestration using LangGraph.
- Experimented with Model Context Protocol (MCP) integration patterns for connecting LLM-based workflows to external tools and structured context.

Technologies: LangGraph, AI Agents, Tool Calling, Model Context Protocol (MCP)

TechPulse — AI Content Automation (Personal Project)

- Built a personal project automating technology-news aggregation and summarization using LLM APIs, exploring practical content-automation workflows.

Technologies: Python, LLM APIs, Content Automation

ADDITIONAL EXPERIENCE

Founder — Sruthi Music Academy (Apr 2025 – Present)

- Founded and operate a Carnatic violin education academy, building its curriculum, fee structure and a Supabase/PostgreSQL-backed student management system with authentication, role-based access and a RAG-based chatbot for student assistance.

EDUCATION

M.Sc. Artificial Intelligence — Department of Computer Science, University of Kerala, 2022

Master of Computer Applications (MCA) — IGNOU, 2018

B.Sc. Computer Science — College of Applied Science (IHRD), Dhanuvachapuram, University of Kerala, 2015

CERTIFICATIONS

Google IT Automation with Python Specialization | Google Data Analytics Specialization